

Common AI application areas

COMPLETE LEARNING GUIDE

What You'll Learn

AI concepts become concrete once you connect them to the application domains they actually power in products you likely already use every day. This submodule surveys the major AI application areas — natural language processing, computer vision, recommendation systems, speech recognition, and generative AI — explaining what problem each area solves, the general approach used, and grounding each in specific, recognizable real-world examples. The goal isn't to make you an expert in any one of these areas (each is itself a large field covered partially in later modules), but to build a mental map so that when you encounter a new AI product or headline, you can quickly place it in the right category and have a reasonable sense of how it likely works and what its likely limitations are.

1. Natural Language Processing (NLP)

NLP covers any AI task involving human language — text or speech — including understanding meaning, generating text, and translating between languages. Real examples include machine translation (Google Translate converting text between languages), sentiment analysis (a company automatically classifying customer reviews or support tickets as positive/negative/neutral to prioritize response), chatbots and virtual assistants (answering questions or routing support requests), and modern large language models (systems like ChatGPT or Claude that generate fluent, contextually relevant text in response to a prompt). The common thread is processing or producing language in a way that captures meaning, not just matching literal keywords.

2. Computer Vision

Computer vision covers AI tasks involving understanding images or video. Image classification labels an entire image with a category (is this a photo of a cat or a dog); object detection goes further, locating and labeling multiple objects within a single image with bounding boxes (used in self-driving cars to identify pedestrians, other vehicles, and traffic signs simultaneously); facial recognition matches a face against a database of known identities (used in phone unlock features and, more controversially, in surveillance); and medical imaging analysis assists radiologists by flagging suspicious regions in X-rays, MRIs, or CT scans for closer human review. Each of these builds on the same core idea — learning to recognize visual patterns from large sets of labeled images — applied to a different specific task.

3. Recommendation Systems

Recommendation systems predict which items a specific user is likely to want, and power a huge share of e-commerce and media revenue. Collaborative filtering recommends items based on the behavior of similar users ('people who bought this also bought...'), working purely from patterns of past interactions without needing to understand the items' content at all. Content-based filtering instead recommends items similar in content/attributes to ones a user already liked (recommending action movies to someone who rated other action movies highly), based on item features rather than other users' behavior. Most production systems (Netflix, Amazon, Spotify) blend both approaches, along with additional signals like recency and popularity, into a hybrid system.

- Collaborative filtering: 'users like you also liked...' — needs no understanding of the item itself, only patterns of co-occurrence across many users.
- Content-based filtering: 'this is similar in content to things you already liked' — needs item features (genre, cast, description) but works even for brand-new users with no interaction history yet.
- The 'cold start' problem — how do you recommend anything to a brand-new user or for a brand-new item with no interaction history? — is a well-known, still only partially solved challenge in this field.

4. Speech Recognition and Generation

Speech recognition converts spoken audio into text (powering voice assistants like Siri and Alexa, live captioning, and voice-to-text dictation), while speech synthesis (text-to-speech) does the reverse, generating natural-sounding spoken audio from text (powering audiobook narration, accessibility tools for visually impaired users, and voice assistant responses). Both have improved dramatically with deep learning, moving from robotic, clearly synthetic-sounding output a decade ago to today's systems that are often difficult to distinguish from a human speaker — which itself raises new concerns around voice-based impersonation and fraud, a theme picked up in the responsible-AI section.

5. Generative AI

Generative AI produces new content — text, images, audio, code, or video — rather than just classifying or predicting a label for existing content. Large language models generate text (this course's own written materials could in principle be drafted with one); diffusion models generate images from text descriptions (tools like Midjourney or DALL-E); and code-generation models assist or automate software writing (GitHub Copilot). Unlike classification or regression, there's no single 'correct' output to compare against — evaluating generative AI quality is a genuinely harder and more subjective problem, typically relying on a combination of automated metrics and human judgment.

6. Anomaly Detection and Fraud Prevention

Anomaly detection identifies data points that deviate significantly from expected patterns, and underlies fraud detection (flagging a credit card transaction that's unusual for a given account's typical spending pattern), network security (detecting unusual traffic patterns that might indicate an intrusion), and industrial equipment monitoring (flagging sensor readings that suggest a machine is about to fail before it actually does). Unlike most of the other application areas above, anomaly detection often has to work with very few or even zero labeled examples of the anomalous case (since, by definition, the interesting failures are rare), which pushes many real systems toward unsupervised or semi-supervised approaches rather than standard supervised classification.

Common Pitfalls

- Assuming every AI application uses the same underlying technique — NLP, computer vision, and recommendation systems typically rely on quite different model architectures suited to their specific data type.
- Treating a chatbot's fluent, confident-sounding text as evidence it 'understands' the conversation the way a human would, rather than as a language-pattern-generation capability.
- Ignoring the cold-start problem when evaluating a recommendation system on paper — performance for brand-new users/items is often much worse than average performance suggests.
- Assuming generative AI output has been fact-checked or verified — generative models can produce fluent, confident, and incorrect content ('hallucination') with no built-in signal that anything is wrong.
- Expecting anomaly detection systems trained on historical fraud/failure patterns to reliably catch entirely novel types of fraud or failure they've never seen anything like before.

Key Takeaways

- NLP, computer vision, recommendation systems, speech, generative AI, and anomaly detection are the major application areas — each solves a different type of problem with different typical approaches.
- Collaborative filtering uses other users' behavior; content-based filtering uses item attributes — most real recommender systems blend both.
- Generative AI's lack of a single 'correct answer' makes evaluating its output quality a fundamentally harder problem than classification or regression.
- Anomaly/fraud detection often has to work with very few labeled examples of the thing it's trying to catch, pushing toward unsupervised approaches.
- Recognizing which application area a new AI product belongs to gives you a reasonable first guess at how it works and where its limitations likely lie.

Learn More

Stanford — Natural Language Processing with Deep Learning (CS224N) Overview

<https://web.stanford.edu/class/cs224n/>

IBM — What Are Recommendation Systems?

<https://www.ibm.com/topics/recommendation-engine>

Google Developers — Machine Learning Crash Course: Computer Vision

<https://developers.google.com/machine-learning/crash-course>